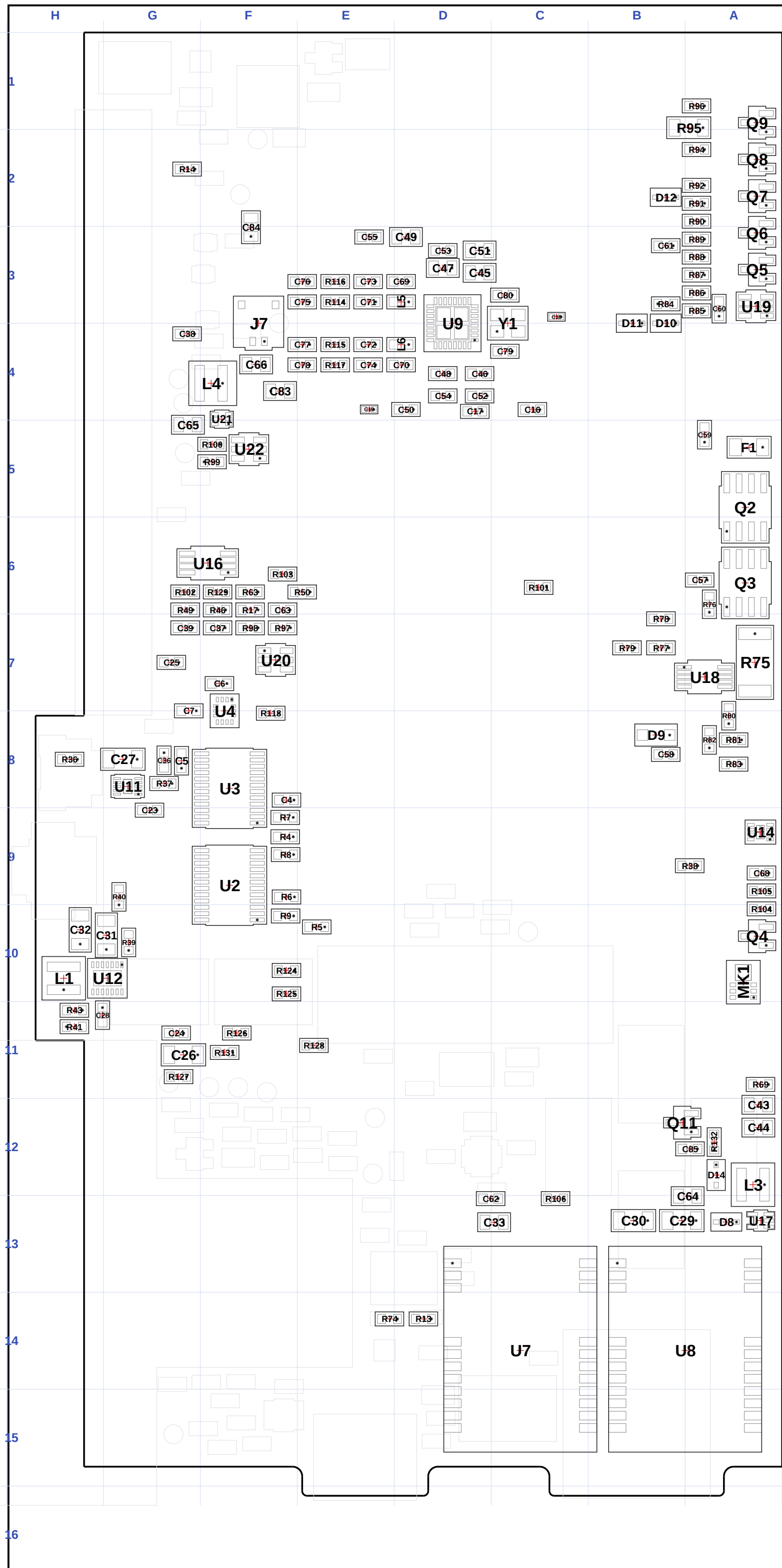




AQROOT DEMO

REFERENCE LOCATOR - BOTTOM

Release D-806

Board sha256:
0e59fb64fe72b0a54b9fee96c19bcae
c2db52a82a871bcec270e064d7fda547

VIEW: BOTTOM (B.Cu), MIRRORED - AS THE ASSEMBLER SEES IT. Grid letters run R->L.

Scale 2.5 : 1 on A3. Grid = 10 mm board.

168 FITTED parts on this side; every one is labelled AT ITS CPL CENTROID (red +).

Placement file: aqroot-Demo-pos-fitted.csv

(JLCPCB copy: *_JLCPCB_CPL.csv).

Index: last page(s) of this file and
agroot-Demo-assembly-ref-index.csv.

LEGEND

dark outline = courtyard (this side)

grey boxes = copper lands

black dot = pad 1 / A1

red + = CPL centroid

faint outline = parts on the other side

CPL frame: X right; Y UP (negative on this board). Board/grid frame here: Y_down = -CPL Y. Bottom-side Rot is as seen from the TOP (KiCad).

DNP - NOT DRAWN, DO NOT PLACE:

C21 C22 C34 C35 C81 C82

L2 R44 R45 R68 R93 R107

R112 R119 R123 U13

Silkscreen intentionally carries no reference designators. This sheet is the authoritative reference locator.

**PARTS PLACEMENT CONFIRMATION IS A
REQUIRED MANUAL APPROVAL GATE.**

REFERENCE INDEX - BOTTOM - D-806 (page 1 of 3)

Ref	Side	Grid	CPL X mm	CPL Y mm	Rot	Value	Package
C4	bottom	F8	51.122820	-79.207791	0	100nF	C_0603_1608Metric
C5	bottom	G8	61.950000	-75.150000	90	100nF	C_0603_1608Metric
C6	bottom	F7	58.041915	-67.186743	0	100nF	C_0603_1608Metric
C7	bottom	G8	61.200000	-70.000000	0	100nF	C_0603_1608Metric
C16	bottom	C4	25.753920	-38.911811	0	100nF	C_0603_1608Metric
C17	bottom	D4	31.700000	-39.100000	0	100nF	C_0603_1608Metric
C18	bottom	C3	23.273385	-29.341214	0	100nF	C_0402_1005Metric
C19	bottom	E4	42.600000	-38.900000	0	100nF	C_0402_1005Metric
C23	bottom	G9	65.250000	-80.250000	0	1uF 25V X7R	C_0603_1608Metric
C24	bottom	G11	62.500000	-103.250000	0	10uF 25V X5R	C_0603_1608Metric
C25	bottom	G7	63.000000	-65.000000	0	1uF 10V X7R	C_0603_1608Metric
C26	bottom	G11	61.750000	-105.500000	0	10uF 10V X7R	C_1206_3216Metric
C27	bottom	G8	68.000000	-75.000000	0	10uF 10V X7R	C_1206_3216Metric
C28	bottom	H11	70.100000	-101.400000	-90	100nF X7R	C_0603_1608Metric
C29	bottom	B13	10.345000	-122.605000	0	22uF 16V X7R	C_1206_3216Metric
C30	bottom	B13	15.335000	-122.605000	0	22uF 16V X7R	C_1206_3216Metric
C31	bottom	G10	69.700000	-93.150000	90	22uF 16V X7R	C_1206_3216Metric
C32	bottom	H10	72.400000	-92.600000	90	22uF 16V X7R	C_1206_3216Metric
C33	bottom	C13	29.705000	-122.775000	0	10uF 10V X7R	C_0805_2012Metric
C36	bottom	G8	63.750000	-75.100000	-90	100nF X7R	C_0603_1608Metric
C37	bottom	F7	58.225000	-61.435000	0	1uF 10V X7R	C_0603_1608Metric
C38	bottom	G4	61.375000	-31.095000	0	1uF 10V X7R	C_0603_1608Metric
C39	bottom	G7	61.575000	-61.435000	0	100nF X7R	C_0603_1608Metric
C43	bottom	A12	2.445000	-110.655000	0	4.7uF 10V X5R	C_0805_2012Metric
C44	bottom	A12	2.445000	-113.005000	0	1uF	C_0805_2012Metric
C45	bottom	D3	31.200000	-24.800000	0	2.2uF	C_0805_2012Metric
C46	bottom	D4	31.200000	-35.200000	0	10nF	C_0603_1608Metric
C47	bottom	D3	35.000000	-24.300000	0	2.2uF	C_0805_2012Metric
C48	bottom	D4	35.000000	-35.200000	0	10nF	C_0603_1608Metric
C49	bottom	D3	38.800000	-21.100000	0	2.2uF	C_0805_2012Metric
C50	bottom	D4	38.800000	-38.900000	0	10nF	C_0603_1608Metric
C51	bottom	D3	31.200000	-22.500000	0	2.2uF	C_0805_2012Metric
C52	bottom	D4	31.200000	-37.500000	0	1nF	C_0603_1608Metric
C53	bottom	D3	35.000000	-22.500000	0	1uF	C_0603_1608Metric
C54	bottom	D4	35.000000	-37.500000	0	10nF	C_0603_1608Metric
C55	bottom	E3	42.600000	-21.100000	0	2.2uF	C_0603_1608Metric
C57	bottom	A6	8.500000	-56.500000	0	4.7nF 50V X7R	C_0603_1608Metric
C58	bottom	B8	12.000000	-74.500000	0	1uF 25V X7R	C_0603_1608Metric
C59	bottom	A5	8.000000	-41.500000	90	1uF 25V X7R	C_0603_1608Metric
C60	bottom	A3	6.500000	-28.500000	90	100nF X7R	C_0603_1608Metric
C61	bottom	B3	12.000000	-22.000000	0	100nF X7R	C_0603_1608Metric
C62	bottom	D13	30.065000	-120.335000	0	1uF 10V X7R	C_0603_1608Metric
C63	bottom	F6	51.525000	-59.585000	0	1uF 10V X7R	C_0603_1608Metric
C64	bottom	A13	9.745000	-120.085000	0	10uF 10V X7R	C_0805_2012Metric
C65	bottom	G5	61.285000	-40.475000	0	22uF 10V X7R	C_0805_2012Metric
C66	bottom	F4	54.250000	-34.250000	0	22uF 10V X7R	C_0805_2012Metric
C68	bottom	A9	2.125000	-86.735000	0	100nF X7R	C_0603_1608Metric
C69	bottom	D3	39.300000	-25.700000	0	100pF 50V C0G TUNE	C_0603_1608Metric
C70	bottom	D4	39.300000	-34.300000	0	100pF 50V C0G TUNE	C_0603_1608Metric
C71	bottom	E3	42.700000	-27.800000	0	300pF 50V C0G TUNE	C_0603_1608Metric
C72	bottom	E4	42.700000	-32.200000	0	300pF 50V C0G TUNE	C_0603_1608Metric
C73	bottom	E3	42.700000	-25.700000	0	1.5nF 50V C0G TUNE	C_0603_1608Metric
C74	bottom	E4	42.700000	-34.300000	0	1.5nF 50V C0G TUNE	C_0603_1608Metric
C75	bottom	E3	49.500000	-27.800000	0	27pF 50V C0G TUNE	C_0603_1608Metric
C76	bottom	E3	49.500000	-25.700000	0	620pF 50V C0G TUNE	C_0603_1608Metric
C77	bottom	E4	49.500000	-32.200000	0	27pF 50V C0G TUNE	C_0603_1608Metric
C78	bottom	E4	49.500000	-34.300000	0	620pF 50V C0G TUNE	C_0603_1608Metric
C79	bottom	C4	28.600000	-32.900000	0	10pF 50V C0G TUNE	C_0603_1608Metric
C80	bottom	C3	28.600000	-27.100000	0	10pF 50V C0G TUNE	C_0603_1608Metric
C83	bottom	F4	51.788000	-37.000000	180	10uF 10V X7R	C_0805_2012Metric
C84	bottom	F3	54.787000	-20.100000	90	10uF 10V X7R	C_0805_2012Metric
C85	bottom	A12	9.500000	-115.200000	0	1uF 10V X7R	C_0603_1608Metric
D8	bottom	A13	5.750000	-122.750000	0	NSR0240	D_SOD-323
D9	bottom	B8	13.000000	-72.500000	0	PMEG2010AEH	D_SOD-123F
D10	bottom	B4	12.000000	-30.000000	0	BAT54WS	D_SOD-323
D11	bottom	B4	15.500000	-30.000000	0	BAT54WS	D_SOD-323
D12	bottom	B2	12.000000	-17.000000	0	BAT54WS	D_SOD-323
D14	bottom	A12	6.800000	-117.900000	-90	BAT54WS	D_SOD-323
F1	bottom	A5	3.400000	-42.800000	0	5A one-shot 1206	Fuse_1206_3216Metric
J7	bottom	F4	54.000000	-30.000000	0	BM02B-ACHSS-GAN-ETF	JST_ACH_BM02B-ACHSS-GAN-ETF_1x02-1MP_P1.
L1	bottom	H10	74.100000	-97.600000	90	1.5uH	Coilcraft_XFL4020
L3	bottom	A12	3.000000	-118.900000	0	4.7uH	Coilcraft_XFL4020
L4	bottom	F4	58.900000	-36.200000	0	1uH	Würth_WE-MAPI_4030_74438357010
L5	bottom	D3	39.300000	-27.800000	0	39nH TUNE	L_0603_1608Metric
L6	bottom	D4	39.300000	-32.200000	0	39nH TUNE	L_0603_1608Metric
MK1	bottom	A10	4.000000	-98.000000	0	DMM-4026-B-I2S	PUI_DMM-4026-B-I2S_4.0x3.0mm
Q2	bottom	A5	3.800000	-49.000000	90	AO4800	SOIC-8_3.9x4.9mm_P1.27mm
Q3	bottom	A6	3.800000	-56.800000	90	AO4800	SOIC-8_3.9x4.9mm_P1.27mm
Q4	bottom	A10	2.575000	-93.255000	0	BSS138	SOT-23
Q5	bottom	A3	2.575000	-24.465000	0	AO3401A	SOT-23
Q6	bottom	A3	2.575000	-20.675000	0	BSS138	SOT-23
Q7	bottom	A2	2.575000	-16.885000	0	BSS138	SOT-23

REFERENCE INDEX - BOTTOM - D-806 (page 2 of 3)

Ref	Side	Grid	CPL X mm	CPL Y mm	Rot	Value	Package
Q8	bottom	A2	2.575000	-13.095000	0	BSS138	SOT-23
Q9	bottom	A1	2.575000	-9.305000	0	BSS138	SOT-23
Q11	bottom	B12	10.300000	-112.500000	0	SQ2364EES-T1_BE3	SOT-23
R4	bottom	F9	51.250000	-83.000000	0	10k	R_0603_1608Metric
R5	bottom	E10	47.999820	-92.300208	0	10k	R_0603_1608Metric
R6	bottom	F9	51.122820	-89.207791	0	10k	R_0603_1608Metric
R7	bottom	F9	51.250000	-81.000000	0	10k	R_0603_1608Metric
R8	bottom	F9	51.210594	-84.830964	0	10k	R_0603_1608Metric
R9	bottom	F10	51.210594	-91.169035	0	10k	R_0603_1608Metric
R13	bottom	D14	37.000000	-132.750000	0	100k	R_0603_1608Metric
R14	bottom	G2	61.375000	-14.095000	0	100k	R_0603_1608Metric
R17	bottom	F6	54.875000	-59.585000	0	100k	R_0603_1608Metric
R36	bottom	H8	73.500000	-75.000000	0	13k 1%	R_0603_1608Metric
R37	bottom	G8	63.750000	-77.500000	0	390R 1%	R_0603_1608Metric
R38	bottom	A9	9.525000	-85.985000	0	10k 1%	R_0603_1608Metric
R39	bottom	G10	67.400000	-93.900000	90	1M 0.1%	R_0603_1608Metric
R40	bottom	G9	68.400000	-89.200000	90	189K 0.1%	R_0603_1608Metric
R41	bottom	H11	73.000000	-102.600000	180	1M	R_0603_1608Metric
R43	bottom	H11	73.000000	-100.900000	0	100K	R_0603_1608Metric
R46	bottom	F6	58.225000	-59.585000	0	10k 1%	R_0603_1608Metric
R49	bottom	G6	61.575000	-59.585000	0	1.5k 1%	R_0603_1608Metric
R50	bottom	E6	49.500000	-57.735000	0	1.5k 1%	R_0603_1608Metric
R63	bottom	F6	54.875000	-57.735000	0	10k 1%	R_0603_1608Metric
R69	bottom	A11	2.225000	-108.555000	0	1.87R 1%	R_0603_1608Metric
R74	bottom	E14	40.500000	-132.750000	0	100k	R_0603_1608Metric
R75	bottom	A7	2.800000	-65.000000	-90	10mR 1% 3W	R_2512_6332Metric
R76	bottom	A6	7.500000	-59.000000	90	22k	R_0603_1608Metric
R77	bottom	B7	12.500000	-63.500000	0	3.65M 1%	R_0603_1608Metric
R78	bottom	B7	12.500000	-60.500000	0	442k 1%	R_0603_1608Metric
R79	bottom	B7	16.000000	-63.500000	0	510k	R_0603_1608Metric
R80	bottom	A8	5.500000	-70.500000	90	1M	R_0603_1608Metric
R81	bottom	A8	5.000000	-73.000000	0	100k	R_0603_1608Metric
R82	bottom	A8	7.500000	-73.000000	90	1M	R_0603_1608Metric
R83	bottom	A8	5.000000	-75.500000	0	10k	R_0603_1608Metric
R84	bottom	B3	12.000000	-28.000000	180	100R	R_0603_1608Metric
R85	bottom	A3	8.825000	-28.725000	0	2.2M 1%	R_0603_1608Metric
R86	bottom	A3	8.825000	-26.875000	0	2.2M 1%	R_0603_1608Metric
R87	bottom	A3	8.825000	-25.025000	0	2.2M 1%	R_0603_1608Metric
R88	bottom	A3	8.825000	-23.175000	0	2.2M 1%	R_0603_1608Metric
R89	bottom	A3	8.825000	-21.325000	0	2.2M 1%	R_0603_1608Metric
R90	bottom	A2	8.825000	-19.475000	0	2.2M 1%	R_0603_1608Metric
R91	bottom	A2	8.825000	-17.625000	0	3.65M 1%	R_0603_1608Metric
R92	bottom	A2	8.825000	-15.775000	0	1.30M 1%	R_0603_1608Metric
R94	bottom	A2	8.825000	-12.075000	0	1M	R_0603_1608Metric
R95	bottom	A1	9.625000	-9.825000	0	560R 1% 1206	R_1206_3216Metric
R96	bottom	A1	8.825000	-7.575000	0	1M	R_0603_1608Metric
R97	bottom	F7	51.525000	-61.435000	0	1.87k 1%	R_0603_1608Metric
R98	bottom	F7	54.875000	-61.435000	0	100k	R_0603_1608Metric
R99	bottom	F5	58.800000	-44.300000	180	732k 1%	R_0603_1608Metric
R100	bottom	F5	58.800000	-42.500000	0	100k 1%	R_0603_1608Metric
R101	bottom	C6	25.107189	-57.258894	0	2.43k 1%	R_0603_1608Metric
R102	bottom	G6	61.575000	-57.735000	0	100k	R_0603_1608Metric
R103	bottom	F6	51.525000	-55.885000	0	100k	R_0603_1608Metric
R104	bottom	A10	2.125000	-90.435000	0	150k 1%	R_0603_1608Metric
R105	bottom	A9	2.125000	-88.585000	0	220k 1%	R_0603_1608Metric
R106	bottom	C13	23.365000	-120.335000	0	0R FIT	R_0603_1608Metric
R114	bottom	E3	46.100000	-27.800000	0	1R1 1% TUNE	R_0603_1608Metric
R115	bottom	E4	46.100000	-32.200000	0	1R1 1% TUNE	R_0603_1608Metric
R116	bottom	E3	46.100000	-25.700000	0	1k TUNE	R_0603_1608Metric
R117	bottom	E4	46.100000	-34.300000	0	1k TUNE	R_0603_1608Metric
R118	bottom	F8	52.750000	-70.250000	0	0R	R_0603_1608Metric
R124	bottom	F10	51.122820	-96.792208	0	1k	R_0603_1608Metric
R125	bottom	F10	51.122820	-99.207791	0	680R	R_0603_1608Metric
R126	bottom	F11	56.250000	-103.250000	0	390R	R_0603_1608Metric
R127	bottom	G11	62.250000	-107.750000	0	10k	R_0603_1608Metric
R128	bottom	E11	48.293772	-104.529969	0	10k	R_0603_1608Metric
R129	bottom	F6	58.225000	-57.735000	0	100k	R_0603_1608Metric
R131	bottom	F11	57.500000	-105.250000	0	100k	R_0603_1608Metric
R132	bottom	A12	7.000000	-114.500000	90	220k 1%	R_0603_1608Metric
U2	bottom	F9	57.000000	-88.000000	0	PCAL9535APW	TSSOP-24_4.4x7.8mm_P0.65mm
U3	bottom	F8	57.000000	-78.000000	0	PCAL9535APW	TSSOP-24_4.4x7.8mm_P0.65mm
U4	bottom	F8	57.500000	-70.000000	-90	BMI270	Bosch_LGA-14_2.5x3.0mm_P0.5mm_BMI270
U7	bottom	C14	27.000000	-136.000000	0	E07-400M10S	Ebyte_E07-400M10S
U8	bottom	B14	10.000000	-136.000000	0	E22-900M22S	Ebyte_E22-900M22S
U9	bottom	D4	34.000000	-30.000000	0	ST25R3916-AQET	ST25R3916_AQET
U11	bottom	G8	67.500000	-77.800000	0	BQ25185	Texas_DLH0010A_WSON-10-1EP_2.2x2mm_P0.4m
U12	bottom	G10	69.600000	-97.600000	0	TPS63020	TI_TPS63020_DSJ
U14	bottom	A9	2.235000	-82.515000	0	MAX17048	MAX17048_T822
U16	bottom	F6	59.250000	-54.750000	0	TCA4307DGKR	VSSOP-8_3x3mm_P0.65mm
U17	bottom	A13	2.200000	-122.600000	0	TPS61169DCKR	SOT-353_SC-70-5
U18	bottom	A7	8.000000	-66.500000	180	LTC4368-1	MSOP-10_3x3mm_P0.5mm
U19	bottom	A3	2.695000	-28.255000	0	TLV7032	SOT-23-8

REFERENCE INDEX - BOTTOM - D-806 (page 3 of 3)

Ref	Side	Grid	CPL X mm	CPL Y mm	Rot	Value	Package
U20	bottom	F7	52.250000	-64.750000	180	TPS22950-Q1	SOT-23-6
U21	bottom	F4	57.800000	-39.900000	0	TPS61023	SOT-563
U22	bottom	F5	55.000000	-43.000000	0	TPS22950-Q1	SOT-23-6
Y1	bottom	C4	28.300000	-30.000000	180	27.12MHz 10pF 3225-4P	Crystal_SMD_3225-4Pin_3.2x2.5mm